Ayush Pathak

AI/ML Engineer | RAG Systems & LLM Applications

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Experience

AI Engineering Intern | Tiledesk (AI agent platform startup), Remote

May 2026 – Sep 2026

- Mapped the AI Agent platform and its RAG Self-Learning Knowledge Base; recommended a lightweight ingestion layer over a rebuilt AI stack, **avoiding duplicate embedding** and retrieval work.
- Prototyped a WhatsApp-to-Knowledge-Base ingestion feature: received webhook events, parsed group messages, and forwarded transcripts into the existing self-learning pipeline.
- Added service modules and routes to an enterprise codebase, resolved dependency issues, and **validated API payloads** with Postman.
- Proposed trigger-based selective ingestion to prevent low-value knowledge generation; wrote integration documentation and handoff notes and contributed to QA and Help Center review.

Projects

RGPV Study Assistant – Hybrid RAG System | Python, LlamaIndex, Pinecone, Flask, Docker

- Built a hybrid RAG system (Pinecone dense search + BM25, Reciprocal Rank Fusion, sigmoid-calibrated CrossEncoder reranker) serving structured exam-format answers over an **8,100+ chunk corpus**.
- Fixed production-grade bugs: uncalibrated reranker logits, a **BM25 zero-IDF** edge case dropping valid matches, and context snowballing across follow-up turns.
- Wrote **a 49-test pytest regression** suite with mocked Pinecone and LLM clients that caught three real bugs pre-deployment; containerized the app with Docker.

Autonomous AI News Curator & Pipeline | Python, Groq, SQLite, GitHub Actions, Streamlit

- Architected a zero-cost ($0/month) pipeline **aggregating 100+ articles** daily across 12 RSS feeds on a 90-second GitHub Actions cron runner.
- Built a two-stage filter (URL deduplication + keyword gate) that **eliminates 70%** of candidate noise and cuts downstream LLM inference overhead by **75%**.
- Deployed Groq-accelerated LLM scoring (gpt-oss-120b) with structured JSON enforcement and backoff retries over **~25 qualified articles**, sending a ranked top-10 HTML newsletter via automated SMTP.

Policy Knowledgebase RAG System | RAG, LangGraph, Embeddings

- Built a prototype RAG pipeline with automated document ingestion and semantic search over a policy knowledge base; used LangGraph for **multi-step agentic workflows** for multi-turn policy lookups.

Technical Skills

- **AI/ML:** RAG, LLM Agents, LangChain, LlamaIndex, LangGraph, Hybrid Search (Dense + BM25), Reranking, Prompt Engineering, Embeddings, Vector Databases (Pinecone, ChromaDB)
- **Backend:** Python, Node.js, Express, FastAPI, Flask, REST APIs, Webhooks, SQL (basic), Docker, Git/GitHub, pytest, Postman
- **ML/Data:** Scikit-learn, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, Streamlit .
- **LLM Providers:** Google Gemini, Groq

Education

Bachelor of Technology, Computer Science | ITM University, Gwalior, Madhya Pradesh **Oct 2024 – Jul 2028**

- **Python Head, Google Developer Student Clubs (GDSC):** lead Python/ML projects and mentor student developers (Sep 2025 – Present).
- **Review paper:** "Deep Learning-Based Face Detection for Smart Surveillance," a comparative analysis of classical to multi-level CNN detection architectures.

Certifications

- Oracle Cloud Infrastructure (OCI) AI Foundations Associate
- Virtual Internship: Artificial Intelligence and Machine Learning
- Python (HackerRank)